Average Value, Accumulated Change, and Motion

Average value, accumulated change from a rate, net displacement, and total distance, mixed so the integral you set up depends on what is asked

Directions. Every problem on this sheet is an integral, but the integral is not the same one each time. Decide what is being asked *before* you set anything up, then show the setup, the antiderivative, and the evaluation. Give exact values unless a problem says otherwise.

- **Average value** of f on $[a, b]$: $\dfrac{1}{b-a}\displaystyle\int_a^b f\,dt$ — a number in the units of f, not of the integral.
- **Accumulated change** from a rate r: $\displaystyle\int_a^b r\,dt$; a starting amount plus the accumulated change gives the amount at the end.
- **Net displacement** of a particle: $\displaystyle\int_a^b v\,dt$ — signed, so motion backwards cancels motion forwards.
- **Total distance** travelled: $\displaystyle\int_a^b |v|\,dt$ — split the interval at every time the velocity changes sign, then add the absolute values.

Round only where a problem asks you to; otherwise leave answers exact.

1. Find the average value of $f(x) = x^2$ on the interval $[0, 3]$.

Answer: ________________

2. Water flows into a tank at a rate of $r(t) = 4t + 3$ litres per minute, where t is measured in minutes.
How much water enters the tank during the first 5 minutes?

Answer: ________________ L

3. A particle moves along a straight line with velocity $v(t) = 3t^2 - 12$ metres per second, for $0 \leq t \leq 3$ seconds.
Find the net displacement of the particle over this interval.

Answer: ________________ m

4. The particle of problem 3 has the same velocity $v(t) = 3t^2 - 12$ metres per second on $0 \leq t \leq 3$.
Find the *total distance* the particle travels. Show where you split the interval and why.

Answer: ________________ m

5. On a spring day the temperature inside a greenhouse is modelled by $T(t) = 18 + 6\sin\left(\frac{\pi t}{12}\right)$ degrees, where t is the number of hours after 6 a.m.
Find the average temperature inside the greenhouse over the 12 hours from $t = 0$ to $t = 12$. Round to two decimal places.

Answer: ________________ degrees

6. A generator's tank holds 14 litres of fuel at $t = 0$. Fuel is consumed at a rate of $c(t) = 0.4 + 0.1t$ litres per hour for $0 \le t \le 4$ hours.
(a) How much fuel is consumed over the four hours?
(b) How much fuel remains in the tank at $t = 4$? Give this amount on the answer line.

Answer: ________________ L

7. A particle moves along a line with velocity $v(t) = t^2 - 4t + 3$ metres per second, for $0 \le t \le 3$ seconds.
(a) At what times in $[0, 3]$ is the particle at rest?
(b) Find the total distance the particle travels on $[0, 3]$.
(c) Find the particle's average speed on $[0, 3]$, in metres per second, rounded to three decimal places. Write this value on the answer line.

Answer: ________________

8. A cyclist starting from rest has velocity $v(t) = 3\sqrt{t}$ metres per second for $0 \leq t \leq 4$ seconds.
Find the cyclist's average velocity over those 4 seconds.

Answer: ________________ m/s

9. A holding tank contains 20 litres of water at $t = 0$. Water enters the tank at a rate of $10 + 2t$ litres per minute and simultaneously drains out at a rate of $3t$ litres per minute, for $0 \leq t \leq 6$ minutes.
(a) Write and evaluate one integral giving the change in the amount of water in the tank over the six minutes.
(b) How much water is in the tank at $t = 6$? Write this amount on the answer line.

Answer: ________________ L

10. **Synthesis challenge.** A particle moves along a line with velocity $v(t) = 4 - t^2$ metres per second for $0 \leq t \leq 3$ seconds. At $t = 0$ its position is $s(0) = 2$ metres.
(a) Find the particle's net displacement on $[0, 3]$.
(b) Find its position $s(3)$.
(c) Find the total distance it travels on $[0, 3]$, rounded to three decimal places.
(d) Find its average velocity on $[0, 3]$.
(e) The particle's average *speed* on $[0, 3]$ is not equal to your answer to part (d). Explain in a short paragraph why not, and say which of the two is larger for this particle.

Answer: ________________